Discrete Mathematics 68/1 > Week 14: 17 Sep 2025 - Pretest > Pretest

Pretest

Status	Finished
Started	Wednesday, 17 September 2025, 9:01 AM
Completed	Wednesday, 17 September 2025, 10:33 AM
Duration	1 hour 32 mins
Grade	**28.75** out of 30.00 (**95.83**%)

Question 1
Correct
Mark 1.00 out of 1.00
Flag question

Determine how many times the innermost loop will be iterated when the algorithm segment is implemented and run.
หาจำนวนครั้งที่ลูปภายในสุดจะถูกทำซ้ำเมื่อส่วนของอัลกอริธึมนี้ถูกนำไปใช้และรัน

```
for i := 1 to 20
  for j := 1 to 10
    [Statements in body of inner loop.
    None contain branching statements that
    lead outside the loop.]
  next j
next i
```

200

✓

The correct answer is: 200

Your answer is correct.

Question 2
Correct
Mark 1.00 out of 1.00
Flag question

Find the first four terms of the sequence defined by a recursively defined sequence.
หาสี่พจน์แรกของลำดับที่กำหนดโดยลำดับเรียกซ้ำ

$a_k = 5ka_{k-1}^2$, for all integers $k \geq 1$
$a_0 = 4$

$a_1 =$ 80 ✓

The correct answer is: 80

$a_2 =$ 64000 ✓

The correct answer is: 64000

$a_3 =$ 614400000

✓

The correct answer is: 61440000000

Your answer is correct.

Quiz navigation

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29 ⊗	30 ✓					

Show one page at a time
Finish review

Question 3
Correct
Mark 1.00 out of 1.00
Flag question

Knee surgery has a 27% chance of success.
The surgery is performed on 4 patients.

Find the probability that the surgery successful for

exactly 0 patients 0.284 ✓

One possible correct answer is: 0.28398241

at most 3 patients 0.9947 ✓

One possible correct answer is: 0.99468559

at least 1 patients 0.716 ✓

One possible correct answer is: 0.71601759

less thant 1 patients 0.284 ✓

One possible correct answer is: 0.28398241

greater than 3 patients 0.0053 ✓

One possible correct answer is: 0.00531441

Your answer is correct.

Question 4
Correct
Mark 1.00 out of 1.00
Flag question

Consider the following algorithm segment:

```
for i := 1 to 4
   for j := 1 to i
      [Statements in body of inner loop. None contain branching statements that lead outside the loop.]
   next j
next i
```

How many times will the inner loop be iterated when the algorithm is implemented and run?

10 ✓

The correct answer is: 10

Your answer is correct.

Question 5
Correct
Mark 1.00 out of 1.00
Flag question

Find explicit formulas for sequences of the form $a_1, a_2, a_3, ...$ with the initial terms given below.
จงหาสูตรสำหรับลำดับในรูปแบบ $a_1, a_2, a_3, ...$ โดยมีการให้ค่าเริ่มต้นของลำดับดังนี้

$\frac{1}{4}, \frac{2}{9}, \frac{3}{16}, \frac{4}{25}, \frac{5}{36}, \frac{6}{49}$

$a_n =$ n/(n+1)^2

✓

The correct answer is: n / (n + 1)^2

Your answer is correct.

Question 6
Correct
Mark 1.00 out of 1.00
Flag question

```
def function(n):
    count = 0
    i=n
    while i >= 1:
        count += 1
        i = i // 3
    return count
```

จาก Master Theorem for Divide and Conquer
ถ้า $f(n) \in O(n^d)$ เมื่อ $d \geq 0$ ในสมการ $T(n) = aT(n/b) + f(n)$ แล้ว

$$T(n) = \begin{cases} O(n^d) & \text{if } a < b^d \\ O(n^d logn) & \text{if } a = b^d \\ O(n^{log_b a}) & \text{if } a > b^d \end{cases}$$

จงหา

$a =$ 1 ✓

One possible correct answer is: 1

$b =$ 3 ✓

One possible correct answer is: 3

$d =$ 0 ✓

One possible correct answer is: 0

What is the complexity of the program given above?
O(logn) ✓

One possible correct answer is: O(logn)

Your answer is correct.

Question 7
Partially correct
Mark 0.75 out of 1.00
Flag question

```
def Function(n):
    if n <= 1:
        return
    for i in range(0, 2):
        Function(n/3)
```

จาก Master Theorem for Divide and Conquer
ถ้า $f(n) \in O(n^d)$ เมื่อ $d \geq 0$ ในสมการ $T(n) = aT(n/b) + f(n)$ แล้ว

$$T(n) = \begin{cases} O(n^d) & \text{if } a < b^d \\ O(n^d logn) & \text{if } a = b^d \\ O(n^{log_b a}) & \text{if } a > b^d \end{cases}$$

จงหา

$a =$ 2 ✓

One possible correct answer is: 2

$b =$ 3 ✓

One possible correct answer is: 3

$d =$ 0 ✓

One possible correct answer is: 0

What is the complexity of the program given above?

O(nlogn) ⊗

One possible correct answer is: O(logn)

Your answer is partially correct.

You have correctly answered 3 parts of this question.

Question 8
Correct
Mark 1.00 out of 1.00
Flag question

Write a single summation.
เขียนผลรวมในพจน์เดียว

$$5 \cdot \sum_{k=1}^{n} (9k - 8) + \sum_{k=1}^{n} (6 - 7k)$$

$= \sum_{k=1}^{n}$ (38k)-34 ✓

The correct answer is: 38 * k + -34

Your answer is correct.

Question 9
Correct
Mark 1.00 out of 1.00
Flag question

Compute
คำนวณ

$\frac{n!}{(n-1)!} =$ n ✓

The correct answer is: n

Your answer is correct.

Question 10
Correct
Mark 1.00 out of 1.00
Flag question

สมมติว่านักวิจัยพิจารณาว่ายาตัวใหม่
มีโอกาส 50% ในการป้องกันไข้หวัดใหญ่สายพันธุ์หนึ่ง
ถ้าให้ยาแก่ผู้เข้าร่วมการทดลอง 4 คน

จงหาความน่าจะเป็นที่ยาจะมีประสิทธิภาพในการป้องกันไข้หวัดใหญ่เท่ากับ 0 คน

0.0625 ✓

One possible correct answer is: 0.0625

จงหาความน่าจะเป็นที่ยาจะมีประสิทธิภาพในการป้องกันไข้หวัดใหญ่น้อยกว่าหรือเท่ากับ 1 คน

0.3125 ✓

One possible correct answer is: 0.3125

จงหาความน่าจะเป็นที่ยาจะมีประสิทธิภาพในการป้องกันไข้หวัดใหญ่มากกว่าหรือเท่ากับ 2 คน

0.6875 ✓

One possible correct answer is: 0.6875

จงหาความน่าจะเป็นที่ยาจะมีประสิทธิภาพในการป้องกันไข้หวัดใหญ่น้อยกว่า 3 คน

0.6875 ✓

One possible correct answer is: 0.6875

จงหาความน่าจะเป็นที่ยาจะมีประสิทธิภาพในการป้องกันไข้หวัดใหญ่มากกว่า 1 คน

0.6875 ✓

One possible correct answer is: 0.6875

Your answer is correct.

Question 11

Correct

Mark 1.00 out of 1.00

Flag question

Find the first four terms of the sequence defined by a recursively defined sequence.
หาสี่พจน์แรกของลำดับที่กำหนดโดยลำดับเรียกซ้ำ

$a_k = 8a_{k-1} + k$, for all integers $k \geq 2$
$a_1 = 6$

$a_2 =$ 50 ✓

The correct answer is: 50

$a_3 =$ 403 ✓

The correct answer is: 403

$a_4 =$ 3228

✓

The correct answer is: 3228

Your answer is correct.

Question 12

Correct

Mark 1.00 out of 1.00

Flag question

Compute the summation
คำนวณผลรวม

$\sum_{n=1}^{14} (n+1) =$ 119

✓

The correct answer is: 119

Your answer is correct.

Question 13
Correct
Mark 1.00 out of 1.00
Flag question

Find explicit formulas for sequences of the form $a_1, a_2, a_3, ...$ with the initial terms given below.
จงหาสูตรสำหรับลำดับในรูปแบบ $a_1, a_2, a_3, ...$ โดยมีการให้ค่าเริ่มต้นของลำดับดังนี้

$\frac{1}{3}, \frac{4}{9}, \frac{9}{27}, \frac{16}{81}, \frac{25}{243}, \frac{36}{729}$

$a_n =$ n^2/3^n

✓

The correct answer is: n^2 / 3^n

Your answer is correct.

Question 14
Correct
Mark 1.00 out of 1.00
Flag question

```
def power(base, exp):
    """Compute the value base**exp for integer exp."""
    if exp == 0:
        return 1

    partial = power(base, exp // 2)
    result = partial * partial
    if exp % 2 == 1:
        result *= base
    return result
```

จาก Master Theorem for Divide and Conquer
ถ้า $f(n) \in O(n^d)$ เมื่อ $d \geq 0$ ในสมการ $T(n) = aT(n/b) + f(n)$ แล้ว

$$T(n) = \begin{cases} O(n^d) & \text{if } a < b^d \\ O(n^d logn) & \text{if } a = b^d \\ O(n^{log_b a}) & \text{if } a > b^d \end{cases}$$

จงหา

$a =$ 1 ✓

One possible correct answer is: 1

$b =$ 2 ✓

One possible correct answer is: 2

$d =$ 0 ✓

One possible correct answer is: 0

What is the complexity of the program given above?
O(logn) ✓

One possible correct answer is: O(logn)

Your answer is correct.

Question 15
Correct
Mark 1.00 out of 1.00
Flag question

Compute
คำนวณ

$\binom{n}{n-1} =$ n ✓

The correct answer is: n

Your answer is correct.

Question 16
Correct
Mark 1.00 out of 1.00
Flag question

Calculate the first four terms of the sequences defined by the formulae.
คำนวณสี่พจน์แรกของลำดับที่กำหนดด้วยสูตร

$a_i = \lfloor \frac{i}{2} \rfloor \cdot 2$, for all integers $i \geq 0$.

$a_i = \lfloor \frac{i}{2} \rfloor \cdot 2$, สำหรับทุกจำนวนเต็ม $i \geq 0$.

$a_0 =$ 0 ✓

The correct answer is: 0

$a_1 =$ 0 ✓

The correct answer is: 0

$a_2 =$ 2 ✓

The correct answer is: 2

$a_3 =$ 2

✓

The correct answer is: 2

Your answer is correct.

Question 17

Correct

Mark 1.00 out of 1.00

Flag question

A worker is promised a bonus if he can increase his productivity by 2 units a day every day for a period of 210 days. If on day 0 he produces 130 units, how many units must he produce on day 210 to qualify for the bonus?
คนงานคนหนึ่งได้รับการสัญญาว่าจะได้โบนัส หากเขาสามารถเพิ่มผลผลิตของเขาได้ 2 หน่วยต่อวันทุกวัน เป็นระยะเวลา 210 วัน หากในวันเริ่มต้น (วันที่ 0) เขาผลิตได้ 130 หน่วย เขาต้องผลิตได้กี่หน่วยในวันที่ 210 จึงจะมีสิทธิ์ได้รับโบนัส?

550

✓

The correct answer is: 550

Your answer is correct.

Question 18

Correct

Mark 1.00 out of 1.00

Flag question

Let $J_5 = \{0, 1, 2, 3, 4\}$, and define a function $F : J_5 \to J_5$ and $G : J_5 \to J_5$ as follows: For each $x \in J_5, f(x) = ((x+4)^2) mod 5$ and $g(x) = (x^2 + 3x + 1) mod 5$.

x	$f(x)$	$g(x)$
0	1	1
1	0	0
2	1	1
3	4	4
4	4	4

✓

The correct answer is: 1; 0; 1; 4; 4; 1; 0; 1; 4; 4

Your answer is correct.

Question 19
Correct
Mark 1.00 out of 1.00
Flag question

How many positive two-digit integers are multiples of 3?

30 ✓

The correct answer is: 30

What is the probability that a randomly chosen positive two-digit integer is a multiple of 3?

1/3 ✓

The correct answer is: 0.33333333333333

What is the probability that a randomly chosen positive two-digit integer is a multiple of 4?

11/45 ✓

The correct answer is: 0.24444444444444

Your answer is correct.

Question 20
Correct
Mark 1.00 out of 1.00
Flag question

A single pair of rabbits (male and female) is born at the beginning of a year. Assume the following conditions.
- Rabbit pairs are not fertile during their first month of life but thereafter give birth to 4 new male/female pairs at the end of every month.
- No rabbits die.
Let $r_n =$ the number of pairs of rabbits alive at the end of month n, for each integer $n \geq 1$, and let $r_0 = 1$.

กระต่ายหนึ่งคู่ (ตัวผู้และตัวเมีย) เกิดขึ้นในต้นปี สมมุติว่ามีเงื่อนไขต่อไปนี้
- กระต่ายจะไม่สามารถเจริญพันธุ์ได้ในช่วงเดือนแรกของชีวิต แต่หลังจากนั้นจะให้กำเนิดคู่กระต่ายใหม่ 4 คู่ (ตัวผู้และตัวเมีย) ในตอนสิ้นเดือน
- ไม่มีกระต่ายตาย"
ให้ $r_n =$ เป็นจำนวนคู่ของกระต่ายที่มีชีวิตอยู่ในตอนสิ้นเดือน n สำหรับจำนวนเต็มทุกจำนวน $n \geq 1$, และให้ $r_0 = 1$

Compute
คำนวณ

$r_0 =$ 1 ✓

The correct answer is: 1

$r_1 =$ 1 ✓

The correct answer is: 1

$r_2 =$ 5 ✓

The correct answer is: 5

$r_3 =$ 9 ✓

The correct answer is: 9

$r_4 =$ 29 ✓

The correct answer is: 29

$r_5 =$ 65 ✓

The correct answer is: 65

$r_6 =$ 181

✓

The correct answer is: 181

Your answer is correct.

Question 21

Correct

Mark 1.00 out of 1.00

Flag question

You are taking a multiple choice quiz that consists of 5 questions.
Each question has 4 possible answers, one of which is correct.
You randomly guess the answer to each question.

Find the probability of guessing

exactly 1 answers correctly 0.3955 ✓

One possible correct answer is: 0.3955078125

at most2 answers correctly 0.8965 ✓

One possible correct answer is: 0.896484375

at least 4 answers correctly 0.0156 ✓

One possible correct answer is: 0.015625

less than 3 answers correctly 0.8965 ✓

One possible correct answer is: 0.896484375

greater than 3 answers correctly 0.0156 ✓

One possible correct answer is: 0.015625

Your answer is correct.

Question 22
Correct
Mark 1.00 out of 1.00
Flag question

Transform an eqation by making the change of variable.
แปลงสมการโดยการเปลี่ยนตัวแปร

$\sum_{i=3}^{n} \frac{i}{i+n-1}$

$= \sum_{j=2}^{n-1}$ (j+1)/(j+n) ✓

The correct answer is: (j + 1) / (j + n)

Your answer is correct.

Question 23
Correct
Mark 1.00 out of 1.00
Flag question

How many 2-permutations are there of a set of 8 objects?

56 ✓

The correct answer is: 56

Your answer is correct.

Question 24
Correct
Mark 1.00 out of 1.00
Flag question

Write equation using summation or product notation.
เขียนสมการโดยใช้สัญลักษณ์ผลรวม

$1^3 + 2^3 + 3^3 + \ldots + n^3$

$= \sum_{k=1}^{n}$ k^3 ✓

The correct answer is: k^3

Your answer is correct.

Question 25
Correct
Mark 1.00 out of 1.00
Flag question

Compute the product
คำนวณผลคูณ

$\prod_{i=2}^{4} i^2 =$ 576

✓

The correct answer is: 576

Your answer is correct.

Question 26
Correct
Mark 1.00 out of 1.00
Flag question

The formula
$1 + r + r^2 + \ldots + r^n = \frac{r^{n+1}-1}{r-1}$

A sequence is defined recursively. Use iteration to guess an explicit formula for the sequence.
ลำดับถูกกำหนดโดยลำดับเรียกซ้ำ ใช้วิธีการวนซ้ำเพื่อคาดเดาสูตรสำหรับลำดับนี้.

$a_k = 6a_{k-1} + 1$, for all integers $k \geq 2$
$a_1 = 1$

$a_2 =$ 7 ✓

The correct answer is: 7

$a_3 =$ 43 ✓

The correct answer is: 43

$a_4 =$ 259 ✓

The correct answer is: 259

$a_5 =$ 1555 ✓

The correct answer is: 1555

$a_n =$ ((6^n)-1)/5

✓

The correct answer is: (6^n - 1) / (6 - 1)

Your answer is correct.

Question 27
Correct
Mark 1.00 out of 1.00
Flag question

Use the formula for the sum of the first n integers to evaluate the sums.
ใช้สูตรผลรวมของจำนวนเต็ม n ตัวแรก เพื่อหาค่าผลรวม

$$20+21+22+23+...+1500$$

= 1125560

✓

The correct answer is: 1125560

Your answer is correct.

Question 28
Correct
Mark 1.00 out of 1.00
Flag question

The formula
$1+2+3+4+...+n=\frac{n(n+1)}{2}$

If n is an integer and $n \geq 1$, find a formula for the expression
ถ้า n เป็นจำนวนเต็มและ $n \geq 1$จงหาสูตรสำหรับนิพจน์

$$6+8+16+24+32+\cdots+8n$$

= (4n^2)+4n+6

✓

The correct answer is: 8 * n * (n + 1) / 2 + 6

Your answer is correct.

Question 29
Incorrect
Mark 0.00 out of 1.00
Flag question

Calculate the first four terms of the sequences defined by the formulae.
คำนวณสี่พจน์แรกของลำดับที่กำหนดด้วยสูตร

$a_i = \frac{22-i}{22+i}$, for all integers $i \geq 1$.
$a_i = \frac{22-i}{22+i}$, สำหรับทุกจำนวนเต็ม $i \geq 1$

$a_1 =$ 1/4 ✗

The correct answer is: 0.91304347826087

$a_2 =$ 1/16 ✗

The correct answer is: 0.83333333333333

$a_3 =$ 1/64 ✗

The correct answer is: 0.76

$a_4 =$ 1/256

✗

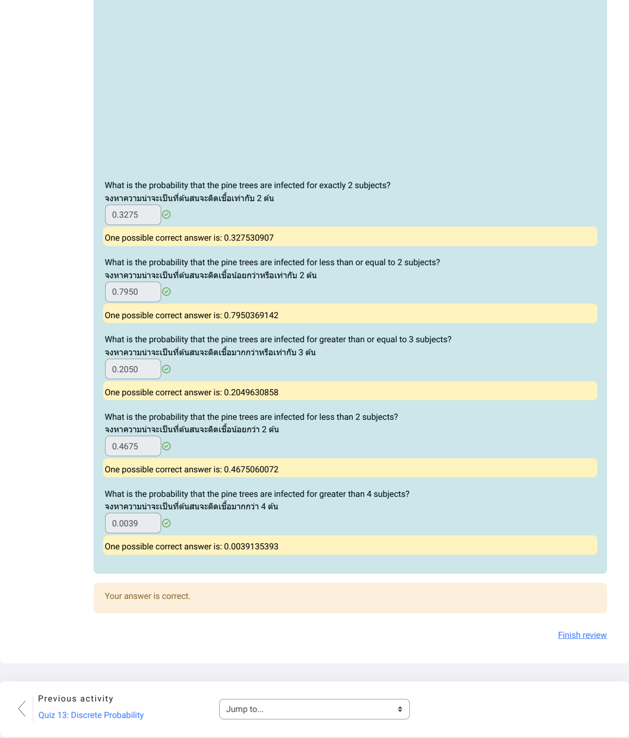
The correct answer is: 0.69230769230769

Your answer is incorrect.

Question 30
Correct
Mark 1.00 out of 1.00
Flag question

Suppose researchers determine that the chances a pine tree in a particular forest is infected by the pine beetle is 33%. If 5 randomly selected pine trees in that forest are tested for infection by the pine beetle,

สมมติว่านักวิจัยพิจารณาว่าโอกาสที่ต้นสนในป่าแห่งหนึ่งจะติดเชื้อจากด้วงสนคือ 33% ถ้าสุ่มเลือกต้นสน 5 ต้นในป่านั้นและทดสอบการติดเชื้อโดยด้วงสน



What is the probability that the pine trees are infected for exactly 2 subjects?
จงหาความน่าจะเป็นที่ต้นสนจะติดเชื้อเท่ากับ 2 ต้น

0.3275 ✓

One possible correct answer is: 0.327530907

What is the probability that the pine trees are infected for less than or equal to 2 subjects?
จงหาความน่าจะเป็นที่ต้นสนจะติดเชื้อน้อยกว่าหรือเท่ากับ 2 ต้น

0.7950 ✓

One possible correct answer is: 0.7950369142

What is the probability that the pine trees are infected for greater than or equal to 3 subjects?
จงหาความน่าจะเป็นที่ต้นสนจะติดเชื้อมากกว่าหรือเท่ากับ 3 ต้น

0.2050 ✓

One possible correct answer is: 0.2049630858

What is the probability that the pine trees are infected for less than 2 subjects?
จงหาความน่าจะเป็นที่ต้นสนจะติดเชื้อน้อยกว่า 2 ต้น

0.4675 ✓

One possible correct answer is: 0.4675060072

What is the probability that the pine trees are infected for greater than 4 subjects?
จงหาความน่าจะเป็นที่ต้นสนจะติดเชื้อมากกว่า 4 ต้น

0.0039 ✓

One possible correct answer is: 0.0039135393

Your answer is correct.

Previous activity
Quiz 13: Discrete Probability

×
Lecture 1 Algorithm Analysis
Video Lecture 1: Algorithm Ana...
Lecture 2 Fundamentals of the ...
Video Lecture 2 Fundamentals ...
Quiz 9: Algorithm Efficiency
Week 11: 27 Aug 2025 - Recur...
Lecture: Recursive
Quiz 10: Recursive
Lecture: Algorithms Design
Quiz 11: Algorithms Complexity
Week 12: 3 Sep 2025 - Counting
Lecture: Counting
Quiz 12: Counting
Week 13: 10 Sep 2025 - Discre...
Lecture: Discrete Probability
Quiz 13: Discrete Probability
Week 14: 17 Sep 2025 - Pretest
Pretest

Discrete Mathematics 68/1 > Week 12: 3 Sep 2025 - Counting > Quiz 12: Counting

Quiz 12: Counting

Status	Finished
Started	Sunday, 21 September 2025, 10:57 PM
Completed	Sunday, 21 September 2025, 11:11 PM
Duration	13 mins 40 secs
Grade	**10.00** out of 10.00 (**100**%)

Question 1
Correct
Mark 1.00 out of 1.00
Flag question

How many positive two-digit integers are multiples of 3?

30 ✓

The correct answer is: 30

What is the probability that a randomly chosen positive two-digit integer is a multiple of 3?

1/3 ✓

The correct answer is: 0.33333333333333

What is the probability that a randomly chosen positive two-digit integer is a multiple of 4?

11/45 ✓

The correct answer is: 0.24444444444444

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 2
Correct
Mark 1.00 out of 1.00
Flag question

Suppose $A[1], A[2], \ldots, A[n]$ is a one-dimensional array and $n \geq 2$. Consider the subarray $A[1], A[2], \ldots, A[n/2]$

สมมติว่า $A[1], A[2], \ldots, A[n]$ เป็นอาร์เรย์หนึ่งมิติและ $n \geq 2$ พิจารณาอาร์เรย์ย่อย $A[1], A[2], \ldots, A[n/2]$

How many elements are in the subarray
มีจำนวนองค์ประกอบกี่ตัวในย่อยของอาร์เรย์

if n is even?
ถ้า n เป็นจำนวนคู่

n/2 ✓

The correct answer is: n / 2

if n is odd?
ถ้า n เป็นจำนวนคี่

(n-1)/2 ✓

The correct answer is: (n - 1) / 2

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 3
Correct
Mark 1.00 out of 1.00
Flag question

Suppose that in a certain state, all automobile license plates have 5 letters followed by 5 digits.
How many different license plates are possible?

1.1881257

✓

The correct answer is: 1188137600000

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 4
Correct
Mark 1.00 out of 1.00
Flag question

How many integers are there from 20 through 69?

50 ✓

The correct answer is: 50

How many odd integers are there from 20 through 69?

25 ✓

The correct answer is: 25

How many integers from 20 through 69 have distinct digits?

45

✓

The correct answer is: 45

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 5
Correct
Mark 1.00 out of 1.00
Flag question

Determine how many times the innermost loop will be iterated when the algorithm segment is implemented and run.
หาจำนวนครั้งที่ลูปภายในสุดจะถูกทำซ้ำเมื่อส่วนของอัลกอริธึมนี้ถูกนำไปใช้และรัน

```
for i := 1 to 10
  for j := 1 to 10
    [Statements in body of inner loop.
    None contain branching statements that
    lead outside the loop.]
  next j
next i
```

100

✓

The correct answer is: 100

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 6
Correct
Mark 1.00 out of 1.00
Flag question

How many ways can the letters of the word ALGORITHM be arranged in a row?

362880 ✓

The correct answer is: 362880

How many ways can the letters of the word ALGORITHM be arranged in a row if A and L must remain together (in order) as a unit?

40320

✓

The correct answer is: 40320

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 7
Correct
Mark 1.00 out of 1.00
Flag question

How many 2-permutations are there of a set of 5 objects?

20 ✓

The correct answer is: 20

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 8
Correct
Mark 1.00 out of 1.00
Flag question

How many ways can three of the letters of the word ALGORITHM be selected and written in a row?

504 ✓

The correct answer is: 504

How many ways can six of the letters of the word ALGORITHM be selected and written in a row if the first letter must be A?

6720 ✓

The correct answer is: 6720

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 9
Correct
Mark 1.00 out of 1.00
Flag question

Consider the following algorithm segment:

```
for i := 1 to 4
  for j := 1 to i
    [Statements in body of inner loop. None contain branching statements that lead outside the loop.]
  next j
next i
```

How many times will the inner loop be iterated when the algorithm is implemented and run?

10 ✓

The correct answer is: 10

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 10
Correct
Mark 1.00 out of 1.00
Flag question

Consider the following algorithm segment:
พิจารณาส่วนของอัลกอริธึมต่อไปนี้:

```
for i := 1 to n
  for j := 1 to i
    [Statements in body of inner loop. None contain branching statements that lead outside the loop.]
  next j
next i
```

How many times will the inner loop be iterated when the algorithm is implemented and run?
ลูปภายในจะถูกทำซ้ำกี่ครั้งเมื่ออัลกอริธึมนี้ถูกนำไปใช้และรัน

n(n+1)/2 ✓

The correct answer is: n * (n + 1) / 2

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Finish review

Previous activity
Lecture: Counting

Jump to...

Next activity
Lecture: Discrete Probability

Quiz 11: Algorithms Complexity

Status	Finished
Started	Sunday, 21 September 2025, 10:48 PM
Completed	Sunday, 21 September 2025, 10:57 PM
Duration	9 mins 11 secs
Marks	9.00/9.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1
Correct
Mark 1.00 out of 1.00
Flag question

Let n be a non-negative integer.
What is the complexity of the program given above?
O(n^3) ⬍ ✓

One possible correct answer is: O(n^3)

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 2
Correct
Mark 1.00 out of 1.00
Flag question

Let n be a non-negative integer.
What is the complexity of the program given above?
O(n) ⬍ ✓

One possible correct answer is: O(n)

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 3
Correct
Mark 1.00 out of 1.00
Flag question

Let n be a non-negative integer.
What is the complexity of the program given above?
O(n) ⬍ ✓

One possible correct answer is: O(n)

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 4
Correct
Mark 1.00 out of 1.00
Flag question

Let n be a non-negative integer.
What is the complexity of the program given above?
O(logn) ⬍ ✓

One possible correct answer is: O(logn)

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 5
Correct
Mark 1.00 out of 1.00
Flag question

```
def function(n):
    count = 0
    i=n
    while i >= 1:
        count += 1
        i = i // 3
    return count
```

จาก Master Theorem for Divide and Conquer
ถ้า $f(n) \in O(n^d)$ เมื่อ $d \geq 0$ ในสมการ $T(n) = aT(n/b) + f(n)$ แล้ว

$$T(n) = \begin{cases} O(n^d) & \text{if } a < b^d \\ O(n^d logn) & \text{if } a = b^d \\ O(n^{log_b a}) & \text{if } a > b^d \end{cases}$$

จงหา

$a =$ 1 ✓

One possible correct answer is: 1

$b =$ 3 ✓

One possible correct answer is: 3

$d =$ 0 ✓

One possible correct answer is: 0

What is the complexity of the program given above?
O(logn) ⬍ ✓

One possible correct answer is: O(logn)

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 6
Correct
Mark 1.00 out of 1.00
Flag question

```
def Function(n):
    count = 0
    if n <= 1:
        return
    for i in range(0, n):
        for j in range(0, n):
            count = count + 1
            print(i,j,count)
    Function(n-1)
```

จาก Master Theorem for Decrease and Conquer

ถ้า $f(n) \in O(n^d)$ เมื่อ $d \geq 0$ ในสมการ

$$T(n) = \begin{cases} c & \text{if } n \leq 1 \\ aT(n-b) + f(n) & \text{if } n > 1 \end{cases}$$

แล้ว

$$T(n) = \begin{cases} O(n^d) & \text{if } a < b^d \\ O(n^{d+1}) & \text{if } a = b^d \\ O(n^d a^{n/b}) & \text{if } a > b^d \end{cases}$$

จงหา

$a =$ 1 ✓

One possible correct answer is: 1

$b =$ 1 ✓

One possible correct answer is: 1

$d =$ 2 ✓

One possible correct answer is: 2

What is the complexity of the program given above?
O(n^3) ⬍ ✓

One possible correct answer is: O(n^3)

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 7
Correct
Mark 1.00 out of 1.00
Flag question

```
def Function(n):
    if n <= 1:
        return
    for i in range(0, 2):
        Function(n/3)
```

จาก Master Theorem for Divide and Conquer
ถ้า $f(n) \in O(n^d)$ เมื่อ $d \geq 0$ ในสมการ $T(n) = aT(n/b) + f(n)$ แล้ว

$$T(n) = \begin{cases} O(n^d) & \text{if } a < b^d \\ O(n^d logn) & \text{if } a = b^d \\ O(n^{log_b a}) & \text{if } a > b^d \end{cases}$$

จงหา

$a =$ 2 ✓

One possible correct answer is: 2

$b =$ 3 ✓

One possible correct answer is: 3

$d =$ 0 ✓

One possible correct answer is: 0

What is the complexity of the program given above?
O(logn) ⬍ ✓

One possible correct answer is: O(logn)

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 8
Correct
Mark 1.00 out of 1.00
Flag question

```
def Function(n):
    count = 0
    if n <= 1:
        return
    for i in range(0, n):
        for j in range(0, n):
            count = count + 1
            print(i,j,count)
    Function(n/2)
```

จาก Master Theorem for Divide and Conquer
ถ้า $f(n) \in O(n^d)$ เมื่อ $d \geq 0$ ในสมการ $T(n) = aT(n/b) + f(n)$ แล้ว

$$T(n) = \begin{cases} O(n^d) & \text{if } a < b^d \\ O(n^d logn) & \text{if } a = b^d \\ O(n^{log_b a}) & \text{if } a > b^d \end{cases}$$

จงหา

$a =$ 1 ✓

One possible correct answer is: 1

$b =$ 2 ✓

One possible correct answer is: 2

$d =$ 2 ✓

One possible correct answer is: 2

What is the complexity of the program given above?
O(n^2) ⬍ ✓

One possible correct answer is: O(n^2)

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 9
Correct
Mark 1.00 out of 1.00
Flag question

```
def power(base, exp):
    """Compute the value base**exp for integer exp."""
    if exp == 0:
        return 1

    partial = power(base, exp // 2)
    result = partial * partial
    if exp % 2 == 1:
        result *= base
    return result
```

จาก Master Theorem for Divide and Conquer
ถ้า $f(n) \in O(n^d)$ เมื่อ $d \geq 0$ ในสมการ $T(n) = aT(n/b) + f(n)$ แล้ว

$$T(n) = \begin{cases} O(n^d) & \text{if } a < b^d \\ O(n^d logn) & \text{if } a = b^d \\ O(n^{log_b a}) & \text{if } a > b^d \end{cases}$$

จงหา

$a =$ 1 ✓

One possible correct answer is: 1

$b =$ 2 ✓

One possible correct answer is: 2

$d =$ 0 ✓

One possible correct answer is: 0

What is the complexity of the program given above?
O(logn) ⬍ ✓

One possible correct answer is: O(logn)

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Finish review

Previous activity
Lecture: Algorithms Design

Jump to... ⬍

Next activity
Lecture: Counting

Quiz 10: Recursive

Status	Finished
Started	Sunday, 21 September 2025, 10:14 PM
Completed	Sunday, 21 September 2025, 10:48 PM
Duration	33 mins 46 secs
Grade	**10.00** out of 10.00 (**100**%)

Question 1
Correct
Mark 1.00 out of 1.00
Flag question

Find the first four terms of the sequence defined by a recursively defined sequence.
หาสี่พจน์แรกของลำดับที่กำหนดโดยลำดับเรียกซ้ำ

$a_k = 5a_{k-1} + k$, for all integers $k \geq 2$
$a_1 = 7$

$a_2 =$ 37 ✓

The correct answer is: 37

$a_3 =$ 188 ✓

The correct answer is: 188

$a_4 =$ 944

✓

The correct answer is: 944

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 2
Correct
Mark 1.00 out of 1.00
Flag question

Find the first four terms of the sequence defined by a recursively defined sequence.
หาสี่พจน์แรกของลำดับที่กำหนดโดยลำดับเรียกซ้ำ

$a_k = 8ka_{k-1}^2$, for all integers $k \geq 1$
$a_0 = 5$

$a_1 =$ 200 ✓

The correct answer is: 200

$a_2 =$ 640000 ✓

The correct answer is: 640000

$a_3 =$ 9.8304*10^

✓

The correct answer is: 9830400000000

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 3
Correct
Mark 1.00 out of 1.00
Flag question

Find the first four terms of the sequence defined by a recursively defined sequence.
หาสี่พจน์แรกของลำดับที่กำหนดโดยลำดับเรียกซ้ำ

$a_k = a_{k-1} + 9a_{k-2}$, for all integers $k \geq 2$
$a_0 = 1, a_1 = 1$

$a_1 =$ 1 ✓

The correct answer is: 1

$a_2 =$ 10 ✓

The correct answer is: 10

$a_3 =$ 19

✓

The correct answer is: 19

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 4
Correct
Mark 1.00 out of 1.00
Flag question

Find the first four terms of the sequence defined by a recursively defined sequence.
หาสี่พจน์แรกของลำดับที่กำหนดโดยลำดับเรียกซ้ำ

$a_k = 4ka_{k-1} - a_{k-2}$, for all integers $k \geq 3$
$a_1 = 1, a_2 = 1$

$a_2 =$ 1 ✓

The correct answer is: 1

$a_3 =$ 11 ✓

The correct answer is: 11

$a_4 =$ 175

✓

The correct answer is: 175

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 5
Correct
Mark 1.00 out of 1.00
Flag question

A single pair of rabbits (male and female) is born at the beginning of a year. Assume the following conditions.
- Rabbit pairs are not fertile during their first month of life but thereafter give birth to 10 new male/female pairs at the end of every month.
- No rabbits die.
Let $r_n =$ the number of pairs of rabbits alive at the end of month n, for each integer $n \geq 1$, and let $r_0 = 1$.

กระต่ายหนึ่งคู่ (ตัวผู้และตัวเมีย) เกิดขึ้นในต้นปี สมมุติว่ามีเงื่อนไขต่อไปนี้
- กระต่ายจะไม่สามารถเจริญพันธุ์ได้ในช่วงเดือนแรกของชีวิต แต่หลังจากนั้นจะให้กำเนิดคู่กระต่ายใหม่ 10 คู่ (ตัวผู้และตัวเมีย) ในตอนสิ้นเดือน
- ไม่มีกระต่ายตาย"
ให้ $r_n =$ เป็นจำนวนคู่ของกระต่ายที่มีชีวิตอยู่ในตอนสิ้นเดือน n สำหรับจำนวนเต็มทุกจำนวน $n \geq 1$, และให้ $r_0 = 1$

Compute
คำนวณ
$r_0 =$ 1 ✓

The correct answer is: 1

$r_1 =$ 1 ✓

The correct answer is: 1

$r_2 =$ 11 ✓

The correct answer is: 11

$r_3 =$ 21 ✓

The correct answer is: 21

$r_4 =$ 131 ✓

The correct answer is: 131

$r_5 =$ 341 ✓

The correct answer is: 341

$r_6 =$ 1651

✓

The correct answer is: 1651

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 6
Correct
Mark 1.00 out of 1.00
Flag question

The formula
$1 + 2 + 3 + 4 + \ldots + n = \frac{n(n+1)}{2}$

If n is an integer and $n \geq 1$, find a formula for the expression
ถ้า n เป็นจำนวนเต็มและ $n \geq 1$ จงหาสูตรสำหรับนิพจน์

$8 + 6 + 12 + 18 + 24 + \cdots + 6n$
= 6n(n+1)/2+8

✓

The correct answer is: 6 * n * (n + 1) / 2 + 8

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 7
Correct
Mark 1.00 out of 1.00
Flag question

The formula
$1 + r + r^2 + \ldots + r^n = \frac{r^{n+1}-1}{r-1}$

If n is an integer and $n \geq 1$, find a formula for the expression
ถ้า n เป็นจำนวนเต็มและ $n \geq 1$ จงหาสูตรสำหรับนิพจน์

$2^n + 2^{n-2} \cdot 7 + 2^{n-3} \cdot 7 + \ldots + 2^2 \cdot 7 + 2 \cdot 7 + 7$
= 9(2^(n-1))-7

✓

The correct answer is: 2^n + 7 * (2^(n - 1) - 1)

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 8
Correct
Mark 1.00 out of 1.00
Flag question

The formula
$1 + r + r^2 + \ldots + r^n = \frac{r^{n+1}-1}{r-1}$

A sequence is defined recursively. Use iteration to guess an explicit formula for the sequence.
ลำดับถูกกำหนดโดยลำดับเรียกซ้ำ ใช้วิธีการวนซ้ำเพื่อคาดเดาสูตรสำหรับลำดับนี้.

$a_k = 5a_{k-1} + 1$, for all integers $k \geq 2$
$a_1 = 1$

$a_2 =$ 6 ✓

The correct answer is: 6

$a_3 =$ 31 ✓

The correct answer is: 31

$a_4 =$ 156 ✓

The correct answer is: 156

$a_5 =$ (156*5)+1 ✓

The correct answer is: 781

$a_n =$ (5^n-1)/4

✓

The correct answer is: (5^n - 1) / (5 - 1)

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 9
Correct
Mark 1.00 out of 1.00
Flag question

The formula
$1 + r + r^2 + \ldots + r^n = \frac{r^{n+1}-1}{r-1}$

A sequence is defined recursively. Use iteration to guess an explicit formula for the sequence.
ลำดับถูกกำหนดโดยลำดับเรียกซ้ำ ใช้วิธีการวนซ้ำเพื่อคาดเดาสูตรสำหรับลำดับนี้.

$a_k = a_{k-1} + 6k$, for all integers $k \geq 1$
$a_0 = 5$

$a_1 =$ 5+6 ✓

The correct answer is: 11

$a_2 =$ 5+6+(6*2) ✓

The correct answer is: 23

$a_3 =$ 5+6+(6*2)+ ✓

The correct answer is: 41

$a_4 =$ 5+6+(6*2)+ ✓

The correct answer is: 65

$a_n =$ (6n^2+6n+10)/2 ✓

The correct answer is: 5 + 6 * n * (n + 1) / 2

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Question 10
Correct
Mark 1.00 out of 1.00
Flag question

A sequence is defined recursively. Use iteration to guess an explicit formula for the sequence.

$a_k = \frac{a_{k-1}}{2a_{k-1}-1}$, for all integers $k \geq 1$
$a_0 = 2$

$a_1 =$ 2/3 ✓

The correct answer is: 0.66666666666667

$a_2 =$ 2 ✓

The correct answer is: 2

$a_3 =$ 2/3 ✓

The correct answer is: 0.66666666666667

$a_4 =$ 2 ✓

The correct answer is: 2

Guess
$a_n =$ 1+1 if n is even
$a_n =$ 2/3 if n is odd

✓

The correct answer is: 2; 0.66666666666667

Your answer is correct.
Correct
Marks for this submission: 1.00/1.00.

Finish review

Previous activity
Lecture: Recursive
Jump to...
Next activity
Lecture: Algorithms Design

Lecture 8: Sequences
Quiz 8 Sequences
Week 10: 20 Aug 2025 - Algori...
Pretest: Algorithm Efficiency
Lecture 1 Algorithm Analysis
Video Lecture 1: Algorithm Ana...
Lecture 2 Fundamentals of the ...
Video Lecture 2 Fundamentals ...
Quiz 9: Algorithm Efficiency
Week 11: 27 Aug 2025 - Recur...
Lecture: Recursive
Quiz 10: Recursive
Lecture: Algorithms Design
Quiz 11: Algorithms Complexity
Week 12: 3 Sep 2025 - Counting
Lecture: Counting
Quiz 12: Counting

Quiz 9: Algorithm Efficiency

Status	Finished
Started	Wednesday, 20 August 2025, 10:02 AM
Completed	Wednesday, 20 August 2025, 10:13 AM
Duration	11 mins 5 secs
Marks	13.00/15.00
Grade	**8.67** out of 10.00 (**86.67**%)

Question 1
Incorrect
Mark 0.00 out of 1.00
Flag question

Let n be a non-negative integer.
What is the complexity of the program given above?

O(nlogn) ✗

One possible correct answer is: O(logn)

Question 2
Correct
Mark 1.00 out of 1.00
Flag question

```
def algorithm(S):
  """Return the sum of the prefix sums of sequence S."""
  n = len(S)
  total = 0
  for j in range(n):            # loop from 0 to n-1
    for k in range(1+j):        # loop from 0 to j
      total += S[k]
  return total
```

What is the complexity of the program given above?

O(n^2) ✓

One possible correct answer is: O(n^2)

Question 3
Correct
Mark 1.00 out of 1.00
Flag question

What is the worst case complexity of the program given above?

O(n) ✓

One possible correct answer is: O(n)

What is the best case complexity of the program given above?

O(1) ✓

One possible correct answer is: O(1)

Question 4
Correct
Mark 1.00 out of 1.00
Flag question

What is the worst case complexity of the program given above?

O(n) ✓

One possible correct answer is: O(n)

What is the best case complexity of the program given above?

O(1) ✓

One possible correct answer is: O(1)

Question 5
Correct
Mark 1.00 out of 1.00
Flag question

What is the worst case complexity of the program given above?

O(n) ✓

One possible correct answer is: O(n)

What is the best case complexity of the program given above?

O(n) ✓

One possible correct answer is: O(n)

Question 6
Correct
Mark 1.00 out of 1.00
Flag question

```
def algorithm(S):
  """Return the sum of the prefix sums of sequence S."""
  n = len(S)
  prefix = 0
  total = 0
  for j in range(n):
    prefix += S[j]
    total += prefix
  return total
```

What is the complexity of the program given above?

O(n) ✓

One possible correct answer is: O(n)

Question 7
Correct
Mark 1.00 out of 1.00
Flag question

```
def function(n):
    total = 0
    for i in range(n):
        total += i
    return total
```

Let n be a non-negative integer.
What is the complexity of the program given above?

O(n) ✓

One possible correct answer is: O(n)

Question 8
Correct
Mark 1.00 out of 1.00
Flag question

```
def algorithm(A, B): # assume that A and B have equal length
  """Return the number of elements in B equal to
  the sum of prefix sums in A."""
  n = len(A)
  count = 0
  for i in range(n):            # loop from 0 to n-1
    total = 0
    for j in range(n):          # loop from 0 to n-1
      for k in range(1+j):      # loop from 0 to j
        total += A[k]
    if B[i] == total:
      count += 1
  return count
```

What is the complexity of the program given above?

O(n^3) ✓

One possible correct answer is: O(n^3)

Question 9
Correct
Mark 1.00 out of 1.00
Flag question

```
def algorithm(S):
  """Return the sum of the elements in sequence S."""
  n = len(S)
  total = 0
  for j in range(n):              # loop from 0 to n-1
    total += S[j]
  return total

```

What is the complexity of the program given above?

O(n) ✓

One possible correct answer is: O(n)

Question 10
Correct
Mark 1.00 out of 1.00
Flag question

```
def function(n):
    count = 0
    for i in range(n):
        count += 1
    for j in range(n):
        count += 1
    return count
```

Let n be a non-negative integer.
What is the complexity of the program given above?

O(n) ✓

One possible correct answer is: O(n)

Question 11
Correct
Mark 1.00 out of 1.00
Flag question

Let n be a non-negative integer.
What is the complexity of the program given above?

O(n) ✓

One possible correct answer is: O(n)

Question 12
Correct
Mark 1.00 out of 1.00
Flag question

Let n be a non-negative integer.
What is the complexity of the program given above?

O(n^2) ✓

One possible correct answer is: O(n^2)

Question 13
Correct
Mark 1.00 out of 1.00
Flag question

Let n be a non-negative integer.
What is the complexity of the program given above?

O(n^2) ✓

One possible correct answer is: O(n^2)

Question 14
Correct
Mark 1.00 out of 1.00
Flag question

```
def algorithm(S):
  """Return the sum of the elements with even index in sequence S.""
  n = len(S)
  total = 0
  for j in range(0, n, 2):       # note the increment of 2
    total += S[j]
  return total
```

What is the complexity of the program given above?

O(n) ✓

One possible correct answer is: O(n)

Question 15
Incorrect
Mark 0.00 out of 1.00
Flag question

Let n be a non-negative integer.
What is the complexity of the program given above?

O(logn) ✗

One possible correct answer is: O(nlogn)

Finish review

Previous activity
Video Lecture 2 Fundamentals of the Analysis of Algorithm Efficiency
Jump to...
Next activity
Lecture: Recursive

Quiz 8 Sequences

Status	Finished
Started	Sunday, 21 September 2025, 9:30 PM
Completed	Sunday, 21 September 2025, 10:07 PM
Duration	37 mins 47 secs
Marks	15.00/15.00
Grade	**10.00** out of 10.00 (**100**%)

Question **1**
Correct
Mark 1.00 out of 1.00
Flag question

Calculate the first four terms of the sequences defined by the formulae.
คำนวณสี่พจน์แรกของลำดับที่กำหนดด้วยสูตร

$a_i = \frac{i}{10+i}$, for all integers $i \geq 1$.
$a_i = \frac{i}{10+i}$, สำหรับทุกจำนวนเต็ม $i \geq 1$

$a_1 =$ 1/11 ✓

The correct answer is: 0.090909090909091

$a_2 =$ 1/6 ✓

The correct answer is: 0.16666666666667

$a_3 =$ 3/13 ✓

The correct answer is: 0.23076923076923

$a_4 =$ 2/7

✓

The correct answer is: 0.28571428571429

Your answer is correct.

Correct
Marks for this submission: 1.00/1.00.

Question **2**
Correct
Mark 1.00 out of 1.00
Flag question

calculate the first four terms of the sequences defined by the formulae.
คำนวณสี่พจน์แรกของลำดับที่กำหนดด้วยสูตร

$a_i = \frac{(-1)^i}{4^i}$, for all integers $i \geq 0$.

$a_i = \frac{(-1)^i}{4^i}$, สำหรับทุกจำนวนเต็ม $i \geq 0$.

$a_0 =$ 1 ✓

The correct answer is: 1

$a_1 =$ -1/4 ✓

The correct answer is: -0.25

$a_2 =$ 1/16 ✓

The correct answer is: 0.0625

$a_3 =$ -1/64

✓

The correct answer is: -0.015625

Your answer is correct.

Correct
Marks for this submission: 1.00/1.00.

Question **3**
Correct
Mark 1.00 out of 1.00
Flag question

Find explicit formulas for sequences of the form $a_1, a_2, a_3, \ldots$ with the initial terms given below.
จงหาสูตรสำหรับลำดับในรูปแบบ $a_1, a_2, a_3, \ldots$ โดยมีการให้ค่าเริ่มต้นของลำดับดังนี้

$-1, 1, -1, 1, -1, 1$

$a_n =$ (-1)^n

✓

The correct answer is: (-1)^n

Your answer is correct.

Correct
Marks for this submission: 1.00/1.00.

Question **4**
Correct
Mark 1.00 out of 1.00
Flag question

Find explicit formulas for sequences of the form $a_1, a_2, a_3, \ldots$ with the initial terms given below.
จงหาสูตรสำหรับลำดับในรูปแบบ $a_1, a_2, a_3, \ldots$ โดยมีการให้ค่าเริ่มต้นของลำดับดังนี้

$0, 1, -2, 3, -4, 5$

$a_n =$ (-1)^n(n-1)

✓

The correct answer is: (n - 1)(-1)^n

Your answer is correct.

Correct
Marks for this submission: 1.00/1.00.

Question **5**
Correct
Mark 1.00 out of 1.00
Flag question

Find explicit formulas for sequences of the form $a_1, a_2, a_3, \ldots$ with the initial terms given below.
จงหาสูตรสำหรับลำดับในรูปแบบ $a_1, a_2, a_3, \ldots$ โดยมีการให้ค่าเริ่มต้นของลำดับดังนี้

$\frac{1}{4}, \frac{2}{9}, \frac{3}{16}, \frac{4}{25}, \frac{5}{36}, \frac{6}{49}$

$a_n =$ n/(n+1)^2

✓

The correct answer is: n / (n + 1)^2

Your answer is correct.

Correct
Marks for this submission: 1.00/1.00.

Question **6**
Correct
Mark 1.00 out of 1.00
Flag question

Find explicit formulas for sequences of the form $a_1, a_2, a_3, \ldots$ with the initial terms given below.
จงหาสูตรสำหรับลำดับในรูปแบบ $a_1, a_2, a_3, \ldots$ โดยมีการให้ค่าเริ่มต้นของลำดับดังนี้

$\frac{1}{3}, \frac{4}{9}, \frac{9}{27}, \frac{16}{81}, \frac{25}{243}, \frac{36}{729}$

$a_n =$ n^2/3^n

✓

The correct answer is: n^2 / 3^n

Your answer is correct.

Correct
Marks for this submission: 1.00/1.00.

Question **7**
Correct
Mark 1.00 out of 1.00
Flag question

Compute the summation
คำนวณผลรวม

$\sum_{n=1}^{54} \left(\frac{1}{n} - \frac{1}{n+1}\right) =$ 54/55

✓

The correct answer is: 0.98148148148148

Your answer is correct.

Correct
Marks for this submission: 1.00/1.00.

Question **8**
Correct
Mark 1.00 out of 1.00
Flag question

Write equation using summation or product notation.
เขียนสมการโดยใช้สัญลักษณ์ผลรวม

$1^2 - 2^2 + 3^2 - 4^2 + 5^2 - 6^2 + 7^2$

$= \sum_{k=1}^{7}$ k^2(-1)^(k+1) ✓

The correct answer is: (-1)^(k + 1) * k^2

Your answer is correct.

Correct
Marks for this submission: 1.00/1.00.

Question **9**
Correct
Mark 1.00 out of 1.00
Flag question

Write equation using summation or product notation.
เขียนสมการโดยใช้สัญลักษณ์ผลรวม

$1 - r + r^2 - r^3 + r^4 - r^5$

$= \sum_{i=0}^{5}$ r^i(-1)^(i) ✓

The correct answer is: (-1)^i * r^i

Your answer is correct.

Correct
Marks for this submission: 1.00/1.00.

Question **10**
Correct
Mark 1.00 out of 1.00
Flag question

Transform an eqation by making the change of variable.
แปลงสมการโดยการเปลี่ยนตัวแปร

$\sum_{k=0}^{5} k(k-1)$

$= \sum_{i=1}^{6}$ (i-1)(i-2) ✓

The correct answer is: (i - 1)(i - 2)

Your answer is correct.

Correct
Marks for this submission: 1.00/1.00.

Question **11**
Correct
Mark 1.00 out of 1.00
Flag question

Transform an eqation by making the change of variable.
แปลงสมการโดยการเปลี่ยนตัวแปร

$\sum_{i=3}^{n} \frac{i}{i+n-1}$

$= \sum_{j=2}^{n-1}$ (j+1)/(j+n) ✓

The correct answer is: (j + 1) / (j + n)

Your answer is correct.

Correct
Marks for this submission: 1.00/1.00.

Question **12**
Correct
Mark 1.00 out of 1.00
Flag question

Use the formula for the sum of the first n integers to evaluate the sums.
ใช้สูตรผลรวมของจำนวนเต็ม n ตัวแรก เพื่อหาค่าผลรวม

$15 + 30 + 45 + 60 + \ldots + 750$

= 19125

✓

The correct answer is: 19125

Your answer is correct.

Correct
Marks for this submission: 1.00/1.00.

Question **13**
Correct
Mark 1.00 out of 1.00
Flag question

Use the formula for the sum of the first n integers to evaluate the sums.
ใช้สูตรผลรวมของจำนวนเต็ม n ตัวแรก เพื่อหาค่าผลรวม

$10 + 11 + 12 + 13 + \ldots + 1900$

= 1805905

✓

The correct answer is: 1805905

Your answer is correct.

Correct
Marks for this submission: 1.00/1.00.

Question **14**
Correct
Mark 1.00 out of 1.00
Flag question

Use the formula for the sum of the first n integers to evaluate the sums.
ใช้สูตรผลรวมของจำนวนเต็ม n ตัวแรก เพื่อหาค่าผลรวม

$1 + 2 + 3 + \ldots + (k-1)$, where k is an integer and $k \geq 2$.

= (k-1)(k)/2 ✓

The correct answer is: k(k - 1) / 2

Your answer is correct.

Correct
Marks for this submission: 1.00/1.00.

Question **15**
Correct
Mark 1.00 out of 1.00
Flag question

Use the formula for the sum of the first n integers and/or the formula for the sum of a geometric sequence to evaluate the sums.

$1 + \frac{1}{2} + \frac{1}{2^2} + \ldots + \frac{1}{2^n}$, where n is a positive integer.

= 2-1/2^n ✓

The correct answer is: 2 - 1 / 2^n

Your answer is correct.

Correct
Marks for this submission: 1.00/1.00.

Finish review

Previous activity
Lecture 8: Sequences

Jump to...

Next activity
Pretest: Algorithm Efficiency

Pretest

Status	Finished
Started	Sunday, 21 September 2025, 11:31 PM
Completed	Sunday, 21 September 2025, 11:54 PM
Duration	23 mins 42 secs
Grade	**26.20** out of 30.00 (**87.33**%)

Question 1
Correct
Mark 1.00 out of 1.00
Flag question

Write equation using summation or product notation.
เขียนสมการโดยใช้สัญลักษณ์ผลรวม

$1^3 + 2^3 + 3^3 + \ldots + n^3$

$= \sum_{k=1}^{n}$ k^3 ✓

The correct answer is: k^3

Your answer is correct.

Question 2
Correct
Mark 1.00 out of 1.00
Flag question

Compute the product
คำนวณผลคูณ

$\prod_{i=2}^{4} i^2 =$ 576

✓

The correct answer is: 576

Your answer is correct.

Question 3
Correct
Mark 1.00 out of 1.00
Flag question

Find explicit formulas for sequences of the form $a_1, a_2, a_3, \ldots$ with the initial terms given below.
จงหาสูตรสำหรับลำดับในรูปแบบ $a_1, a_2, a_3, \ldots$ โดยมีการให้ค่าเริ่มต้นของลำดับดังนี้

$\frac{1}{4}, \frac{2}{9}, \frac{3}{16}, \frac{4}{25}, \frac{5}{36}, \frac{6}{49}$

$a_n =$ n/(n+1)^2

✓

The correct answer is: n / (n + 1)^2

Your answer is correct.

Question 4

Correct

Mark 1.00 out of 1.00

Flag question

Let $J_5 = \{0, 1, 2, 3, 4\}$, and define a function $F : J_5 \to J_5$ and $G : J_5 \to J_5$ as follows: For each $x \in J_5$, $f(x) = ((x+4)^2) mod 5$ and $g(x) = (x^2 + 3x + 1) mod 5$.

x	$f(x)$	$g(x)$
0	1	1
1	0	0
2	1	1
3	4	4
4	4	4

✓

The correct answer is: 1; 0; 1; 4; 4; 1; 0; 1; 4; 4

Your answer is correct.

Question 5

Correct

Mark 1.00 out of 1.00

Flag question

Find the first four terms of the sequence defined by a recursively defined sequence.
หาสี่พจน์แรกของลำดับที่กำหนดโดยลำดับเรียกซ้ำ

$a_k = 8a_{k-1} + k$, for all integers $k \geq 2$
$a_1 = 9$

$a_2 =$ 74 ✓

The correct answer is: 74

$a_3 =$ 595 ✓

The correct answer is: 595

$a_4 =$ 4764

✓

The correct answer is: 4764

Your answer is correct.

Question 6

Correct

Mark 1.00 out of 1.00

Flag question

Consider the following algorithm segment:

```
for i := 1 to 4
   for j := 1 to i
      [Statements in body of inner loop. None contain branching statements that lead outside the loop.]
   next j
next i
```

How many times will the inner loop be iterated when the algorithm is implemented and run?

10 ✓

The correct answer is: 10

Your answer is correct.

Question 7
Correct
Mark 1.00 out of 1.00
Flag question

How many 4-permutations are there of a set of 5 objects?

120 ✓

The correct answer is: 120

Your answer is correct.

Question 8
Correct
Mark 1.00 out of 1.00
Flag question

A worker is promised a bonus if he can increase his productivity by 3 units a day every day for a period of 30 days. If on day 0 he produces 110 units, how many units must he produce on day 30 to qualify for the bonus?
คนงานคนหนึ่งได้รับการสัญญาว่าจะได้โบนัส หากเขาสามารถเพิ่มผลผลิตของเขาได้ 3 หน่วยต่อวันทุกวัน เป็นระยะเวลา 30 วัน หากในวันเริ่มต้น (วันที่ 0) เขาผลิตได้ 110 หน่วย เขาต้องผลิตได้กี่หน่วยในวันที่ 30 จึงจะมีสิทธิ์ได้รับโบนัส?

200

✓

The correct answer is: 200

Your answer is correct.

Question 9
Correct
Mark 1.00 out of 1.00
Flag question

Determine how many times the innermost loop will be iterated when the algorithm segment is implemented and run.
หาจำนวนครั้งที่ลูปภายในสุดจะถูกทำซ้ำเมื่อส่วนของอัลกอริธึมนี้ถูกนำไปใช้และรัน

```
for i := 1 to 10
   for j := 1 to 10
      [Statements in body of inner loop.
      None contain branching statements that
      lead outside the loop.]
   next j
next i
```

100

✓

The correct answer is: 100

Your answer is correct.

Question 10
Correct
Mark 1.00 out of 1.00
Flag question

```python
def function(n):
    count = 0
    i=n
    while i >= 1:
        count += 1
        i = i // 3
    return count
```

จาก Master Theorem for Divide and Conquer
ถ้า $f(n) \in O(n^d)$ เมื่อ $d \geq 0$ ในสมการ $T(n) = aT(n/b) + f(n)$ แล้ว

$$T(n) = \begin{cases} O(n^d) & \text{if } a < b^d \\ O(n^d logn) & \text{if } a = b^d \\ O(n^{log_b a}) & \text{if } a > b^d \end{cases}$$

จงหา

$a =$ 1 ✓

One possible correct answer is: 1

$b =$ 3 ✓

One possible correct answer is: 3

$d =$ 0 ✓

One possible correct answer is: 0

What is the complexity of the program given above?

O(logn) ✓

One possible correct answer is: O(logn)

Your answer is correct.

Question 11
Correct
Mark 1.00 out of 1.00
Flag question

How many positive two-digit integers are multiples of 3?

30 ✓

The correct answer is: 30

What is the probability that a randomly chosen positive two-digit integer is a multiple of 3?

1/3 ✓

The correct answer is: 0.33333333333333

What is the probability that a randomly chosen positive two-digit integer is a multiple of 4?

11/45 ✓

The correct answer is: 0.24444444444444

Your answer is correct.

Question 12
Partially correct
Mark 0.20 out of 1.00
Flag question

You are taking a multiple choice quiz that consists of 5 questions.
Each question has 4 possible answers, one of which is correct.
You randomly guess the answer to each question.

Find the probability of guessing

exactly 0 answers correctly 0.2373046 ✓

One possible correct answer is: 0.2373046875

at most4 answers correctly ⊗

One possible correct answer is: 0.9990234375

at least 4 answers correctly ⊗

One possible correct answer is: 0.015625

less than 3 answers correctly ⊗

One possible correct answer is: 0.896484375

greater than 2 answers correctly ⊗

One possible correct answer is: 0.103515625

Your answer is partially correct.

You have correctly answered 1 part of this question.

Question 13
Not answered
Marked out of 1.00
Flag question

สมมติว่านักวิจัยพิจารณาว่ายาตัวใหม่
มีโอกาส 45% ในการป้องกันไข้หวัดใหญ่สายพันธุ์หนึ่ง
ถ้าให้ยาแก่ผู้เข้าร่วมการทดลอง 4 คน

จงหาความน่าจะเป็นที่ยาจะมีประสิทธิภาพในการป้องกันไข้หวัดใหญ่เท่ากับ 0 คน

⊗

One possible correct answer is: 0.09150625

จงหาความน่าจะเป็นที่ยาจะมีประสิทธิภาพในการป้องกันไข้หวัดใหญ่น้อยกว่าหรือเท่ากับ 3 คน

⊗

One possible correct answer is: 0.95899375

จงหาความน่าจะเป็นที่ยาจะมีประสิทธิภาพในการป้องกันไข้หวัดใหญ่มากกว่าหรือเท่ากับ 2 คน

⊗

One possible correct answer is: 0.60901875

จงหาความน่าจะเป็นที่ยาจะมีประสิทธิภาพในการป้องกันไข้หวัดใหญ่น้อยกว่า 2 คน

⊗

One possible correct answer is: 0.39098125

จงหาความน่าจะเป็นที่ยาจะมีประสิทธิภาพในการป้องกันไข้หวัดใหญ่มากกว่า 2 คน

☐ ⊗

One possible correct answer is: 0.24148125

Your answer is incorrect.

Question **14**

Correct

Mark 1.00 out of 1.00

Flag question

The formula

$1 + r + r^2 + ... + r^n = \frac{r^{n+1}-1}{r-1}$

A sequence is defined recursively. Use iteration to guess an explicit formula for the sequence.
ลำดับถูกกำหนดโดยลำดับเรียกซ้ำ ใช้วิธีการวนซ้ำเพื่อคาดเดาสูตรสำหรับลำดับนี้.

$a_k = 4a_{k-1} + 1,$ for all integers $k \geq 2$
$a_1 = 1$

$a_2 =$ 5 ✓

The correct answer is: 5

$a_3 =$ 21 ✓

The correct answer is: 21

$a_4 =$ 85 ✓

The correct answer is: 85

$a_5 =$ 341 ✓

The correct answer is: 341

$a_n =$ (4^n-1)/3

✓

The correct answer is: (4^n - 1) / (4 - 1)

Your answer is correct.

Question **15**

Correct

Mark 1.00 out of 1.00

Flag question

Calculate the first four terms of the sequences defined by the formulae.
คำนวณสี่พจน์แรกของลำดับที่กำหนดด้วยสูตร

$a_i = \frac{26-i}{26+i}$, for all integers $i \geq 1$.
$a_i = \frac{26-i}{26+i}$, สำหรับทุกจำนวนเต็ม $i \geq 1$

$a_1 =$ 25/27 ✓

The correct answer is: 0.925925925925926

$a_2 =$ 24/28 ✓

The correct answer is: 0.857142857142857

$a_3 =$ 23/29 ✓

The correct answer is: 0.793103448275862

$a_4 =$ 22/30

✓

The correct answer is: 0.73333333333333

Your answer is correct.

Question 16

Correct

Mark 1.00 out of 1.00

Flag question

Use the formula for the sum of the first n integers to evaluate the sums.
ใช้สูตรผลรวมของจำนวนเต็ม n ตัวแรก เพื่อหาค่าผลรวม

$40 + 41 + 42 + 43 + \ldots + 1800$

= 1620120

✓

The correct answer is: 1620120

Your answer is correct.

Question 17

Correct

Mark 1.00 out of 1.00

Flag question

Compute
คำนวณ

$\frac{n!}{(n-1)!} =$ n ✓

The correct answer is: n

Your answer is correct.

Question 18

Correct

Mark 1.00 out of 1.00

Flag question

Find explicit formulas for sequences of the form $a_1, a_2, a_3, \ldots$ with the initial terms given below.
จงหาสูตรสำหรับลำดับในรูปแบบ $a_1, a_2, a_3, \ldots$ โดยมีการให้ค่าเริ่มต้นของลำดับดังนี้

$\frac{1}{3}, \frac{4}{9}, \frac{9}{27}, \frac{16}{81}, \frac{25}{243}, \frac{36}{729}$

$a_n =$ n^2/3^n

✓

The correct answer is: n^2 / 3^n

Your answer is correct.

Question 19
Correct
Mark 1.00 out of 1.00
Flag question

A single pair of rabbits (male and female) is born at the beginning of a year. Assume the following conditions.
- Rabbit pairs are not fertile during their first month of life but thereafter give birth to 6 new male/female pairs at the end of every month.
- No rabbits die.
Let $r_n =$ the number of pairs of rabbits alive at the end of month n, for each integer $n \geq 1$, and let $r_0 = 1$.

กระต่ายหนึ่งคู่ (ตัวผู้และตัวเมีย) เกิดขึ้นในต้นปี สมมุติว่ามีเงื่อนไขต่อไปนี้
- กระต่ายจะไม่สามารถเจริญพันธุ์ได้ในช่วงเดือนแรกของชีวิต แต่หลังจากนั้นจะให้กำเนิดคู่กระต่ายใหม่ 6 คู่ (ตัวผู้และตัวเมีย) ในตอนสิ้นเดือน
- ไม่มีกระต่ายตาย"
ให้ $r_n =$ เป็นจำนวนคู่ของกระต่ายที่มีชีวิตอยู่ในตอนสิ้นเดือน n สำหรับจำนวนเต็มทุกจำนวน $n \geq 1$, และให้ $r_0 = 1$

Compute
คำนวณ

$r_0 =$ 1 ✓

The correct answer is: 1

$r_1 =$ 1 ✓

The correct answer is: 1

$r_2 =$ 7 ✓

The correct answer is: 7

$r_3 =$ 13 ✓

The correct answer is: 13

$r_4 =$ 55 ✓

The correct answer is: 55

$r_5 =$ 78+55 ✓

The correct answer is: 133

$r_6 =$ 78+55+330

✓

The correct answer is: 463

Your answer is correct.

Question 20
Correct
Mark 1.00 out of 1.00
Flag question

Calculate the first four terms of the sequences defined by the formulae.
คำนวณสี่พจน์แรกของลำดับที่กำหนดด้วยสูตร

$a_i = \lfloor \frac{i}{2} \rfloor \cdot 3$, for all integers $i \geq 0$.

$a_i = \lfloor \frac{i}{2} \rfloor \cdot 3$, สำหรับทุกจำนวนเต็ม $i \geq 0$.

$a_0 =$ 0 ✓

The correct answer is: 0

$a_1 =$ 0 ✓

The correct answer is: 0

$a_2 =$ 3 ✓

The correct answer is: 3

$a_3 =$ 3

✓

The correct answer is: 3

Your answer is correct.

Question 21
Correct
Mark 1.00 out of 1.00
Flag question

Write a single summation.
เขียนผลรวมในพจน์เดียว

$9 \cdot \sum_{k=1}^{n} (4k - 8) + \sum_{k=1}^{n} (5 - 8k)$

$= \sum_{k=1}^{n}$ 28k-67 ✓

The correct answer is: 28 * k + -67

Your answer is correct.

Question 22
Correct
Mark 1.00 out of 1.00
Flag question

Compute
คำนวณ

$\binom{n}{n-1} =$ n ✓

The correct answer is: n

Your answer is correct.

Question 23
Correct
Mark 1.00 out of 1.00
Flag question

```
def Function(n):
    if n <= 1:
        return
    for i in range(0, 2):
        Function(n/3)
```

จาก Master Theorem for Divide and Conquer
ถ้า $f(n) \in O(n^d)$ เมื่อ $d \geq 0$ ในสมการ $T(n) = aT(n/b) + f(n)$ แล้ว

$$T(n) = \begin{cases} O(n^d) & \text{if } a < b^d \\ O(n^d logn) & \text{if } a = b^d \\ O(n^{log_b a}) & \text{if } a > b^d \end{cases}$$

จงหา

$a =$ 2 ✓

One possible correct answer is: 2

$b =$ 3 ✓

One possible correct answer is: 3

$d =$ 0 ✓

One possible correct answer is: 0

What is the complexity of the program given above?

O(logn) ✓

One possible correct answer is: O(logn)

Your answer is correct.

Question 24

Not answered

Marked out of 1.00

Flag question

Suppose researchers determine that the chances a pine tree in a particular forest is infected by the pine beetle is 80%. If 5 randomly selected pine trees in that forest are tested for infection by the pine beetle,

สมมติว่านักวิจัยพิจารณาว่าโอกาสที่ต้นสนในป่าแห่งหนึ่งจะติดเชื้อจากด้วงสนคือ 80% ถ้าสุ่มเลือกต้นสน 5 ต้นในป่านั้นและทดสอบการติดเชื้อโดยด้วงสน

What is the probability that the pine trees are infected for exactly 3 subjects?
จงหาความน่าจะเป็นที่ต้นสนจะติดเชื้อเท่ากับ 3 ต้น

⊗

One possible correct answer is: 0.2048

What is the probability that the pine trees are infected for less than or equal to 3 subjects?
จงหาความน่าจะเป็นที่ต้นสนจะติดเชื้อน้อยกว่าหรือเท่ากับ 3 ต้น

⊗

One possible correct answer is: 0.26272

What is the probability that the pine trees are infected for greater than or equal to 4 subjects?
จงหาความน่าจะเป็นที่ต้นสนจะติดเชื้อมากกว่าหรือเท่ากับ 4 ต้น

⊗

One possible correct answer is: 0.73728

What is the probability that the pine trees are infected for less than 2 subjects?
จงหาความน่าจะเป็นที่ต้นสนจะติดเชื้อน้อยกว่า 2 ต้น

⊗

One possible correct answer is: 0.00672

What is the probability that the pine trees are infected for greater than 4 subjects?
จงหาความน่าจะเป็นที่ต้นสนจะติดเชื้อมากกว่า 4 ต้น

⊗

One possible correct answer is: 0.32768

Your answer is incorrect.

Question **25**

Not answered

Marked out of 1.00

Flag question

Knee surgery has a 27% chance of success.
The surgery is performed on 4 patients.

Find the probability that the surgery successful for

exactly 1 patients ⊗

One possible correct answer is: 0.42013836

at most 3 patients ⊗

One possible correct answer is: 0.99468559

at least 2 patients ⊗

One possible correct answer is: 0.29587923

less thant 3 patients ⊗

One possible correct answer is: 0.93721123

greater than 2 patients ⊗

One possible correct answer is: 0.06278877

Your answer is incorrect.

Question **26**

Correct

Mark 1.00 out of 1.00

Flag question

Find the first four terms of the sequence defined by a recursively defined sequence.
หาสี่พจน์แรกของลำดับที่กำหนดโดยลำดับเรียกซ้ำ

$a_k = 7ka_{k-1}^2$, for all integers $k \geq 1$
$a_0 = 5$

$a_1 =$ 175 ✓

The correct answer is: 175

$a_2 =$ 428750 ✓

The correct answer is: 428750

$a_3 =$ 3.86035781

The correct answer is: 3860357812500

Your answer is correct.

Question 27
Correct
Mark 1.00 out of 1.00
Flag question

The formula
$1+2+3+4+...+n=\frac{n(n+1)}{2}$

If n is an integer and $n \geq 1$, find a formula for the expression
ถ้า n เป็นจำนวนเต็มและ $n \geq 1$จงหาสูตรสำหรับนิพจน์

$9+7+14+21+28+\cdots+7n$
= 7n(n+1)/2+9

The correct answer is: 7 * n * (n + 1) / 2 + 9

Your answer is correct.

Question 28
Correct
Mark 1.00 out of 1.00
Flag question

Compute the summation
คำนวณผลรวม

$\sum_{n=1}^{14}(n+1)=$ 119

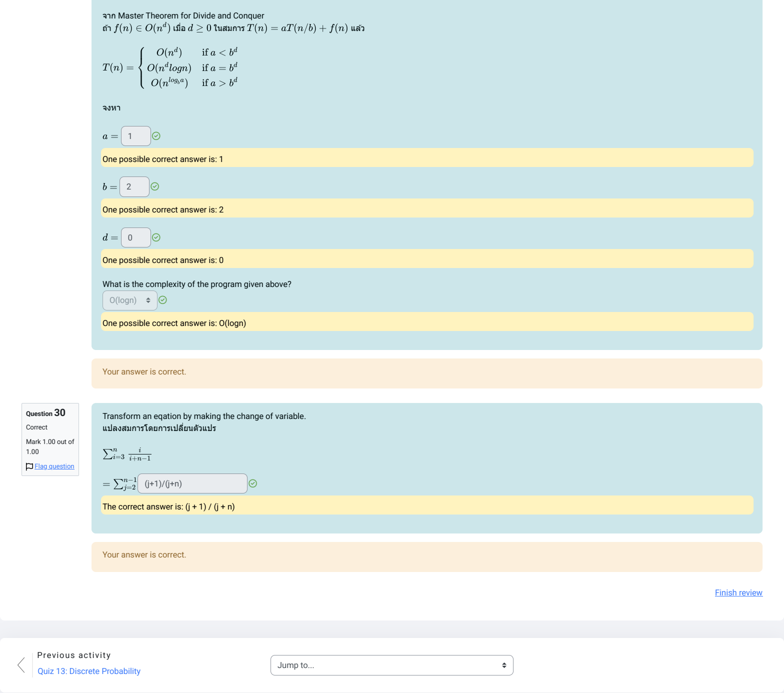
The correct answer is: 119

Your answer is correct.

Question 29
Correct
Mark 1.00 out of 1.00
Flag question

```
def power(base, exp):
    """Compute the value base**exp for integer exp."""
    if exp == 0:
        return 1

    partial = power(base, exp // 2)
    result = partial * partial
    if exp % 2 == 1:
        result *= base
    return result
```



จาก Master Theorem for Divide and Conquer
ถ้า $f(n) \in O(n^d)$ เมื่อ $d \geq 0$ ในสมการ $T(n) = aT(n/b) + f(n)$ แล้ว

$$T(n) = \begin{cases} O(n^d) & \text{if } a < b^d \\ O(n^d logn) & \text{if } a = b^d \\ O(n^{log_b a}) & \text{if } a > b^d \end{cases}$$

จงหา

$a =$ 1 ✓

One possible correct answer is: 1

$b =$ 2 ✓

One possible correct answer is: 2

$d =$ 0 ✓

One possible correct answer is: 0

What is the complexity of the program given above?

O(logn) ✓

One possible correct answer is: O(logn)

Your answer is correct.

Question 30
Correct
Mark 1.00 out of 1.00

Transform an eqation by making the change of variable.
แปลงสมการโดยการเปลี่ยนตัวแปร

$\sum_{i=3}^{n} \frac{i}{i+n-1}$

$= \sum_{j=2}^{n-1}$ (j+1)/(j+n) ✓

The correct answer is: (j + 1) / (j + n)

Your answer is correct.